

CHAPTER 3

SUPPLY AND DEMAND: THEORY



DOWNER/GETTY IMAGES



REG 36 HJK 30 +1/4 ASD 45 -1/3 CROS 10 -5/8 RES 36 HJK 30
 REG 36 HJK 30 +1/4 ASD 45 -1/3 CROS 10 -5/8 RES 36 HJK 30
 REG 36 HJK 30 +1/4 ASD 45 -1/3 CROS 10 -5/8 RES 36 HJK 30

In This Lecture

➤ Demand

➤ Supply

➤ The Market

➤ Consumers' Surplus,
Producers' Surplus, and
Total Surplus



To select a topic, click on its link above

Market

Any place people come together to trade

Trade or exchange
may take place at a
physical or virtual
location



Demand - A Definition

The willingness and ability of buyers:

- to purchase different quantities of a good
- at different prices
- during a specific time period.



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Law of Demand

As the price of a good rises, the quantity demanded of the good falls, and as the price of a good falls, the quantity demanded of the good rises, ceteris paribus

Price



↑
fahle kaitan

nyun

Yan g

cin

Quantity



Negative relationship

Ceteris Paribus

A Latin term meaning “all other things constant” or “nothing else changes.”

Ceteris paribus is an assumption used to examine the effect of one influence on an outcome while holding all other influences Constant.

Quantity Demanded – A Definition

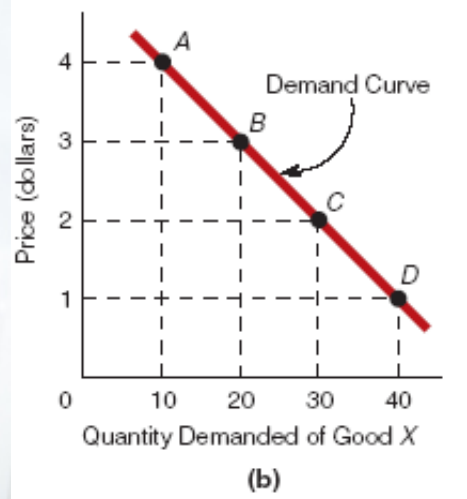
- Quantity Demanded is the number of units (say 100 units of a good) that individuals are willing and able to buy at a particular price (say, \$10 per unit) during a time period.
- This is different from demand which speaks to the willingness and ability of buyers to buy different quantities of a good at different prices.

- satu jumlah kuantiti yg ada willingness
at some specific price

- every price are men at some quantity
based on people

Demand

Click graph to watch tutorial on demand.



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Demand Schedule

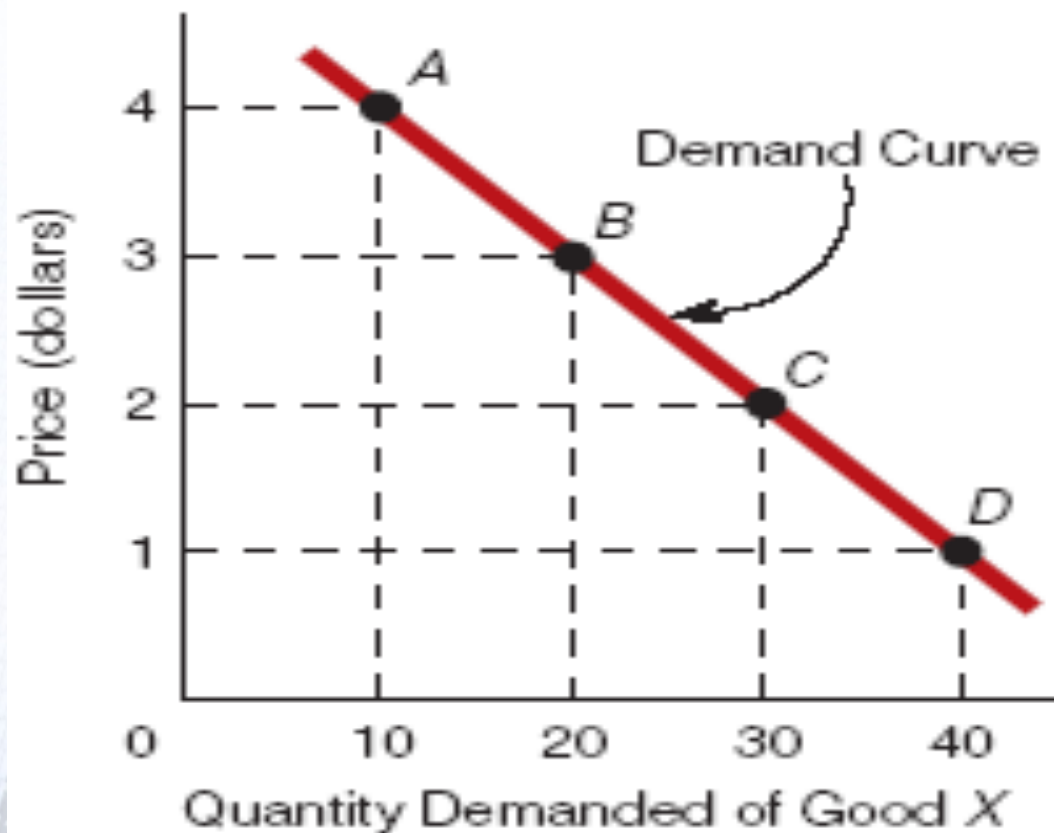
The numerical tabulation of the quantity demanded of a good at different prices.

A demand schedule is the numerical representation of the law of demand.

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Downward Sloping Demand Curve

The graphical representation of the demand schedule and law of demand.

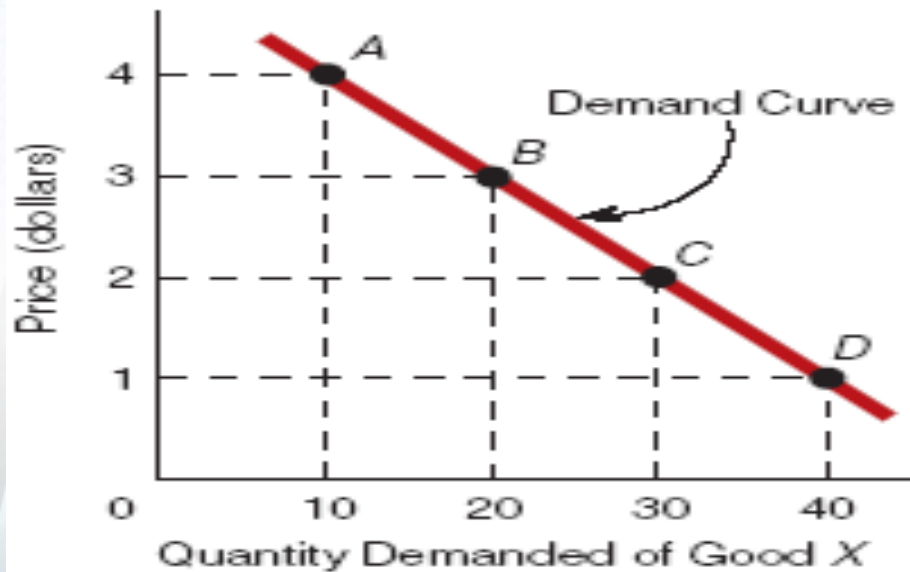


Demand - Schedule and Graph

Demand Schedule for Good X

Price (dollars)	Quantity Demanded	Point in Part (b)
4	10	<i>A</i>
3	20	<i>B</i>
2	30	<i>C</i>
1	40	<i>D</i>

(a)



(b)

Why Does Quantity Demanded Go Down as Price Goes Up?

- people substitute lower priced goods for higher priced goods.
- the law of diminishing marginal utility, which states that for a given time period, the marginal (or additional) utility or satisfaction gained by consuming equal successive units of a good will decline as the amount consumed increases.

Satisfaction ↓, Quantity ↑

- is Nature manusia sendiri

Marginal Utility ↓, Quantity ↑

- orang hanya dapat 1st air, sukanya gile... Kalau 10 air

Why does the price of one more day at Disney World cost less than the cost of the first day?



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Individual Demand Curve vs Market Demand Curve

- An Individual Demand Curve represents the price-quantity combination of a particular single buyer. For example an Individual demand curve could show the Jones's demand for CDs.
- A Market demand curve represents the price-quantity combination of a good for all buyers. In this case, the demand curve would show all buyers demand for CDs.

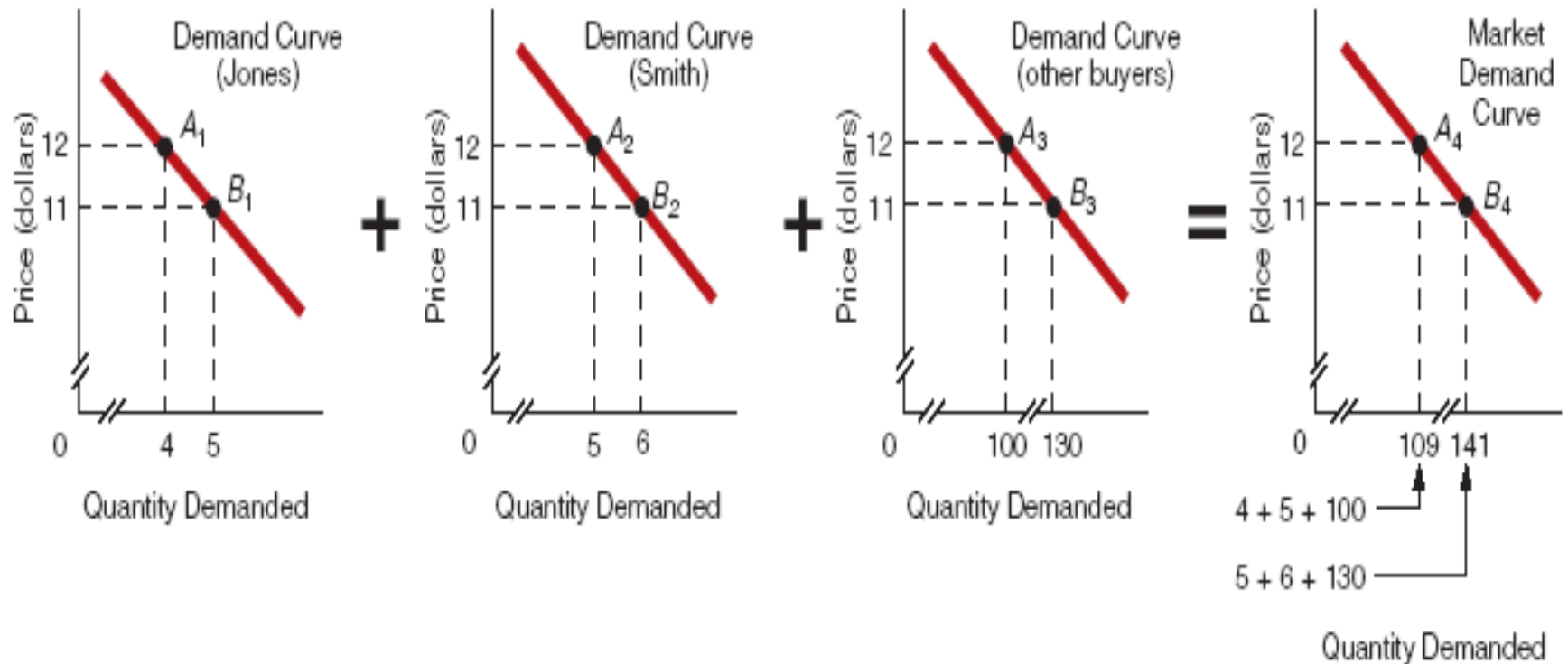
Derivation of Market Demand Schedule

Horizontal submission

Price	Quantity Demanded						
	Jones		Smith		Other Buyers		All Buyers
\$15	1		2		20		23
14	2		3		45		50
13	3		4		70		77
12	4	+	5	+	100	=	109
11	5	+	6	+	130	=	141
10	6		7		160		173

(a)

Derivation of Market Demand Curve



(b)

Increase in Demand

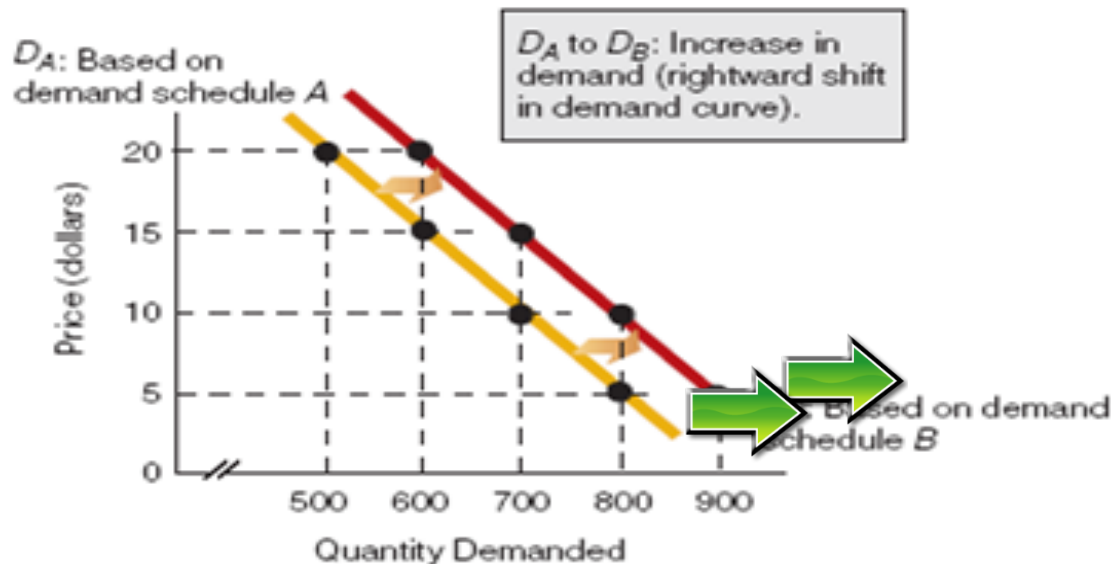
Shift

Demand Schedule A

Price	Quantity Demanded
\$20	500
\$15	600
\$10	700
\$ 5	800

Demand Schedule B (increase in demand)

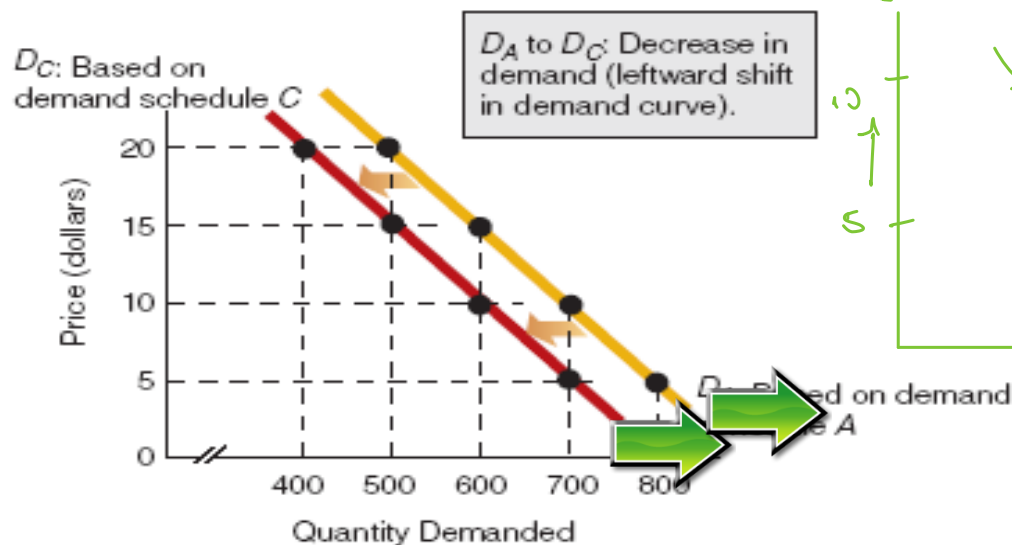
Price	Quantity Demanded
\$20	500 600
\$15	600 700
\$10	700 800
\$ 5	800 900



Decrease in Demand

Demand Schedule A	
Price	Quantity Demanded
\$20	500
\$15	600
\$10	700
\$ 5	800

Demand Schedule C (decrease in demand)	
Price	Quantity Demanded
\$20	500 400
\$15	600 500
\$10	700 600
\$ 5	800 700



(b)

Factors Causing a Shift in the Demand Curve

- Income
- Preferences
- Prices of substitute goods
- Prices of complementary goods
- Number of buyers
- Expectations of future prices

Factors Causing a Shift in the Demand Curve - Income

Normal Good - A good the demand for which rises (falls) as income rises (falls).  Brand

Inferior Good - A good the demand for which falls (rises) as income rises (falls).  Bundle

Neutral Good - A good the demand for which does not change as income rises or falls.  Beer

Preferences

Anggapan ... can ruka foh beli foh
Demand of Tea

Factors Causing a Shift in the Demand Curve – Prices of Related Goods

Substitutes

Two goods that satisfy similar needs or desires. If two goods are substitutes, the demand for one rises as the price of the other rises (or the demand for one falls as the price of the other falls).

Coffee ngan tea

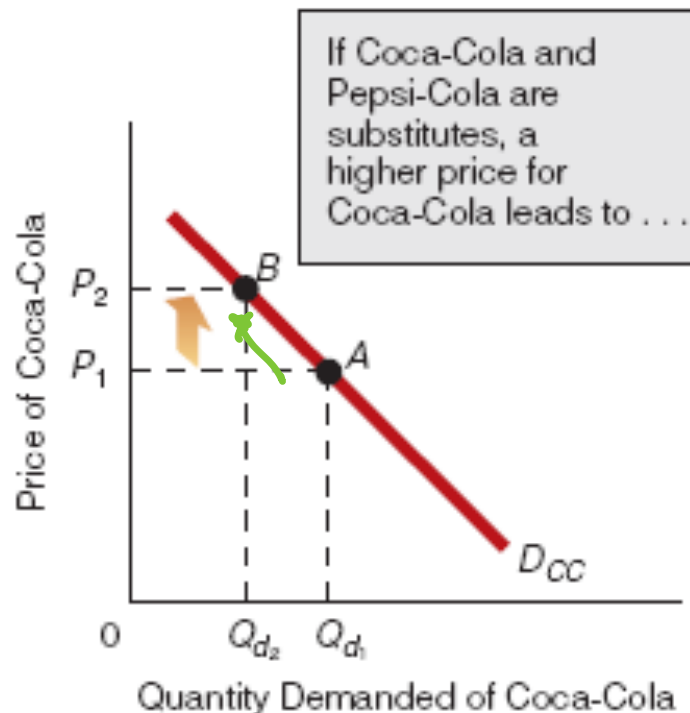
Complements

Two goods that are used jointly in consumption. If two goods are complements, the demand for one rises as the price of the other falls (or the demand for one falls as the price of the other rises).

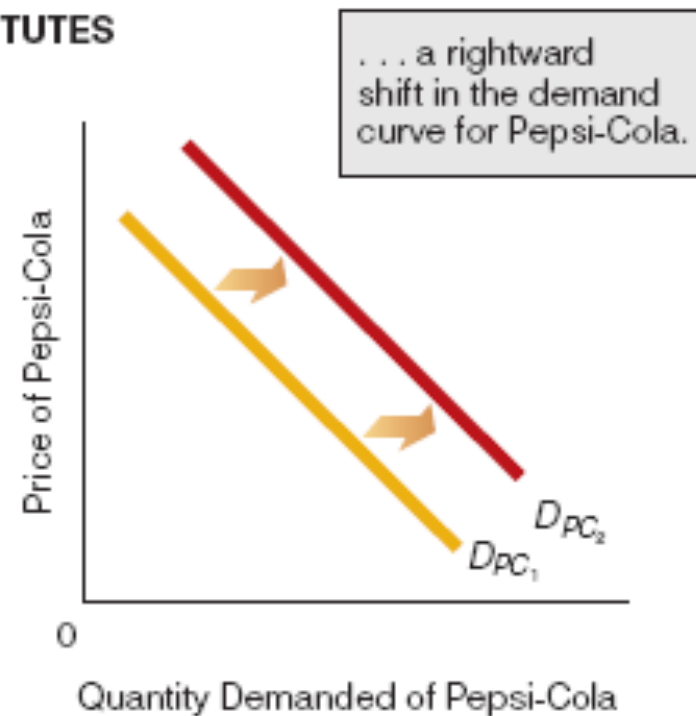
satu set

Printer ngan cartridge

Effect on Demand if Price Increases on a Substitute Good

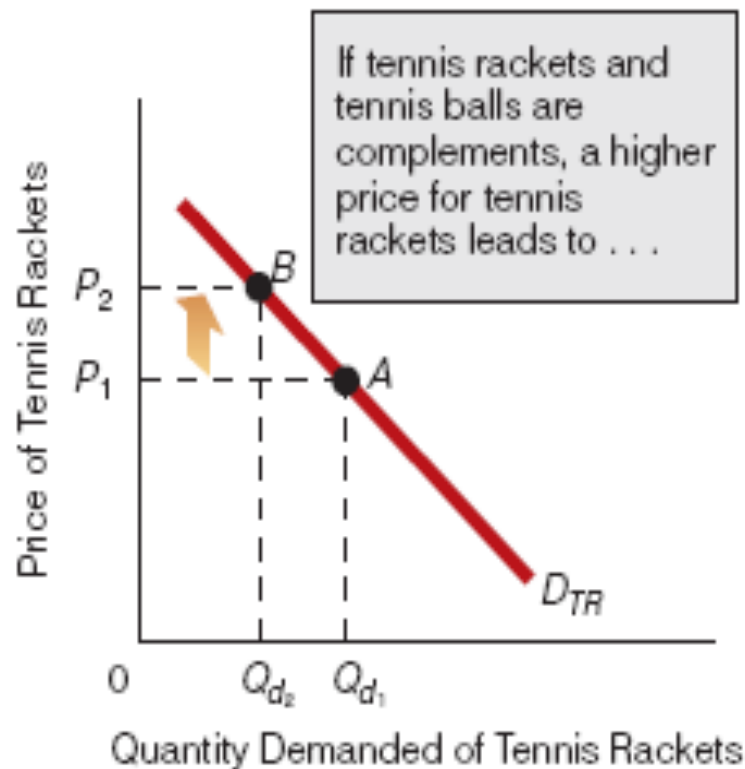


SUBSTITUTES

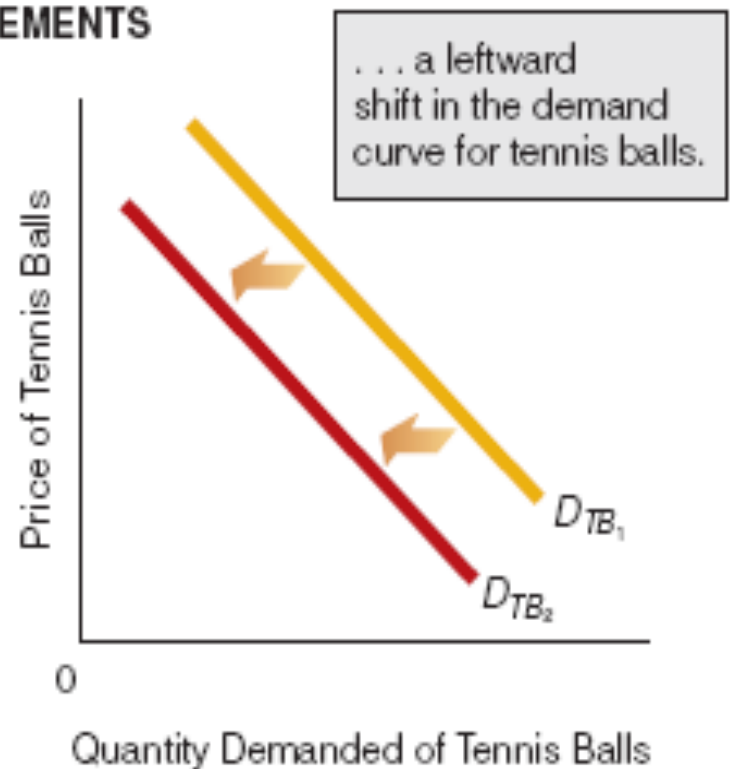


(a)

Effect on Demand if Price Increases on a Complementary Good



COMPLEMENTS



(b)

Factors Causing a Shift in the Demand Curve – Number of Buyers

Number of Buyers

The demand for a good in a particular market area is related to the number of buyers in the area; the more buyers, higher demand; fewer buyers, lower demand. The number of buyers may increase owing to a heightened birthrate, increased immigration, the migration of people from one region of the country to another, and so on. The number of buyers may decrease owing to an increased death rate, war the migration of people from one region of the country to another, and so on.

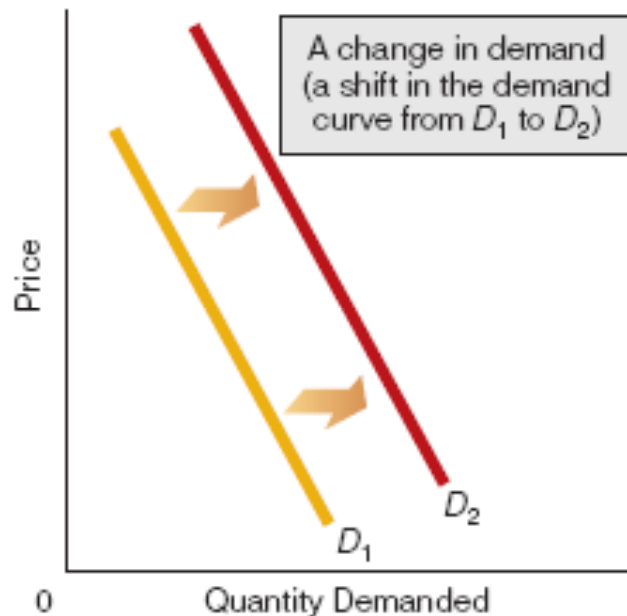
Factors Causing a Shift in the Demand Curve – Expectations of Future Price

Expectations of Future Price

Buyers who expect the price of a good to be higher next month may buy it now, thus increasing the current (or present) demand for the good. Buyers who expect the price of a good to be lower next month may wait until the next month to buy it, thus decreasing the current (or present) demand for the good.

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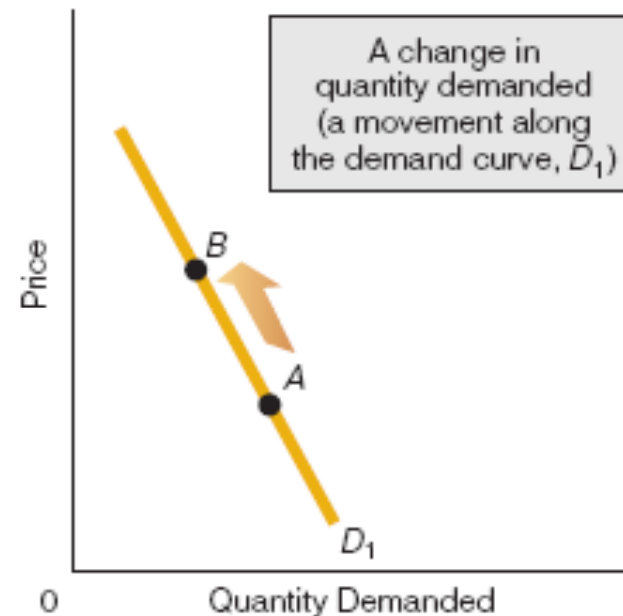
A Change in Demand versus a Change in Quantity Demanded



A change in any of these (shift) factors can cause a change in demand:

1. Income
2. Preferences
3. Prices of related goods
4. Number of buyers
5. Expectations of future price

(a)



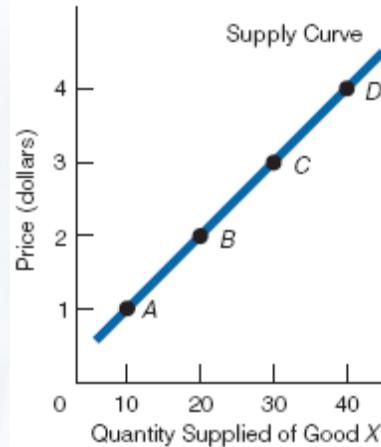
A change in this (movement) factor will cause a change in quantity demanded:

1. (A good's) own price

(b)

Supply

Click graph to watch tutorial



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Supply

The willingness and ability of sellers to produce and offer to sell different quantities of a good at different prices during a specific time period.

Law of Supply

As the price of a good rises, the quantity supplied of the good rises, and as the price of a good falls, the quantity supplied of the good falls, *ceteris paribus*.

Price

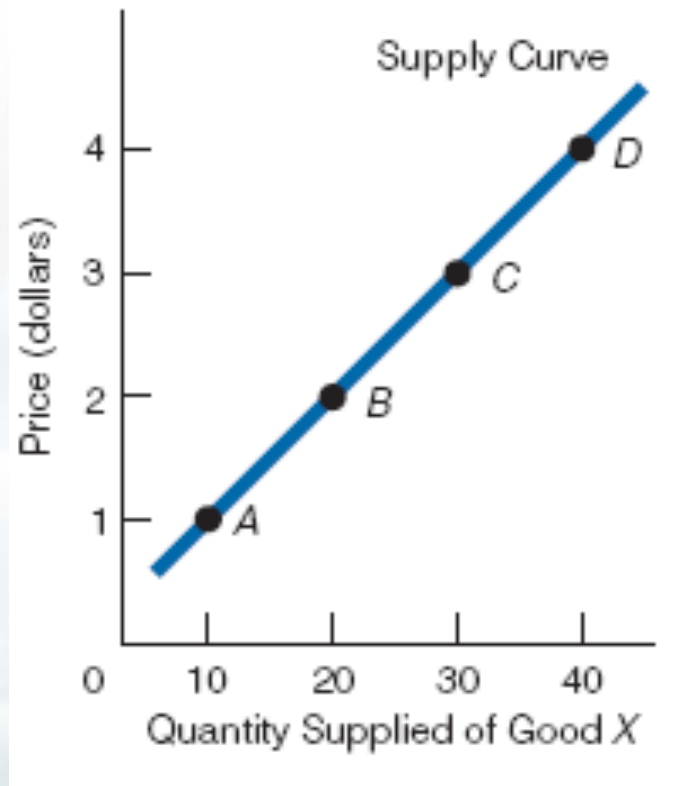


Quantity



Supply Curve

The graphical representation of the law of supply, which states that price and quantity supplied are directly related, *ceteris paribus*.

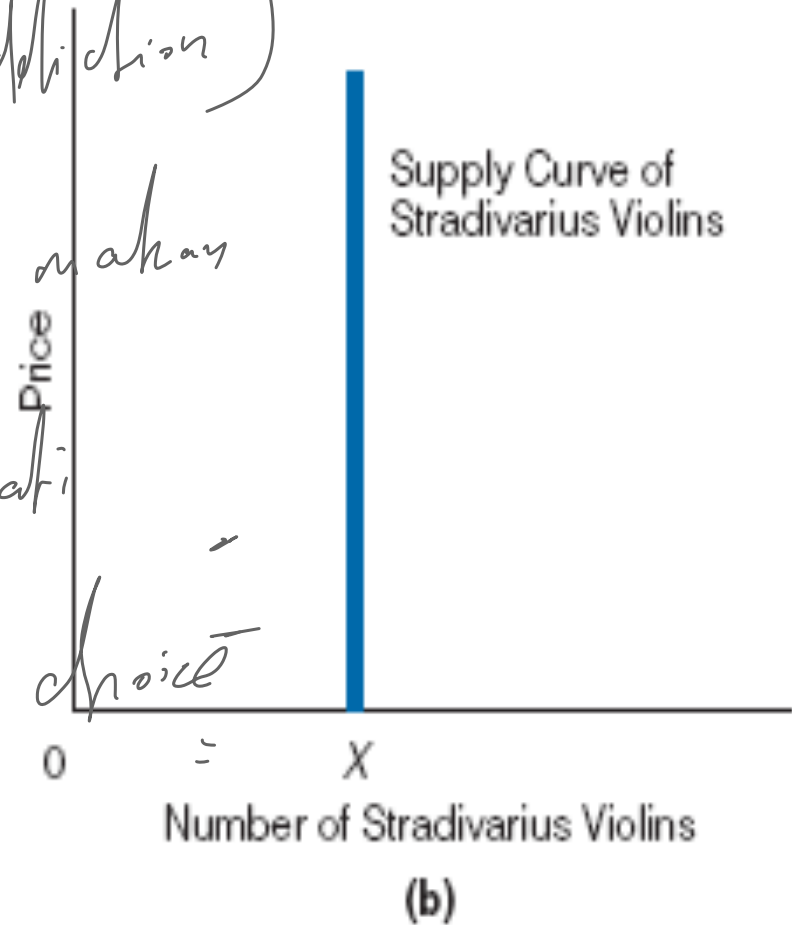
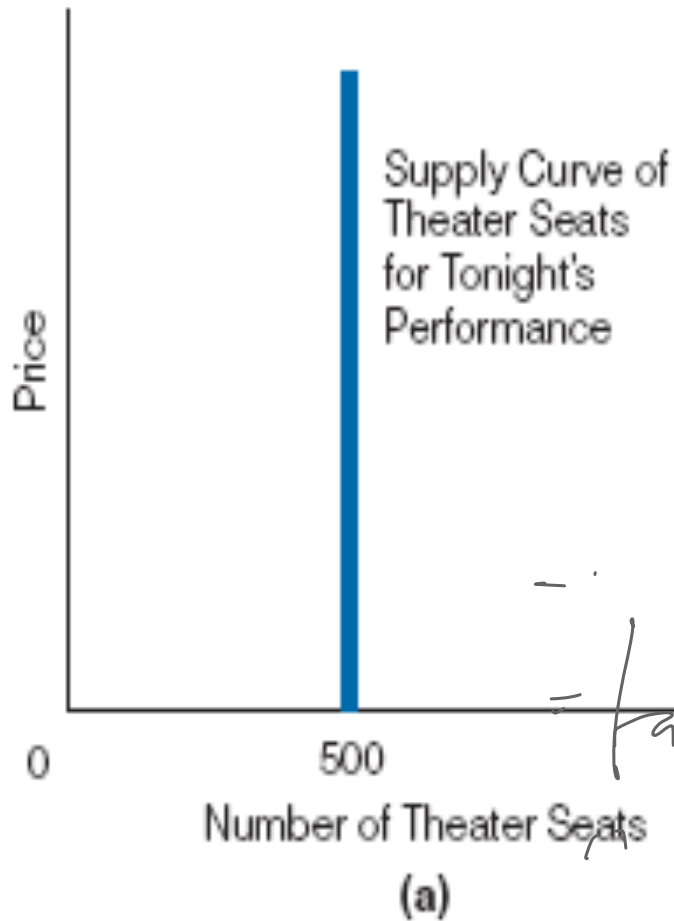


Fixed Supply

Rich, Drugs
(Addiction)

Fah mahay
leh mati

- fahde
- choice



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Supply Schedule

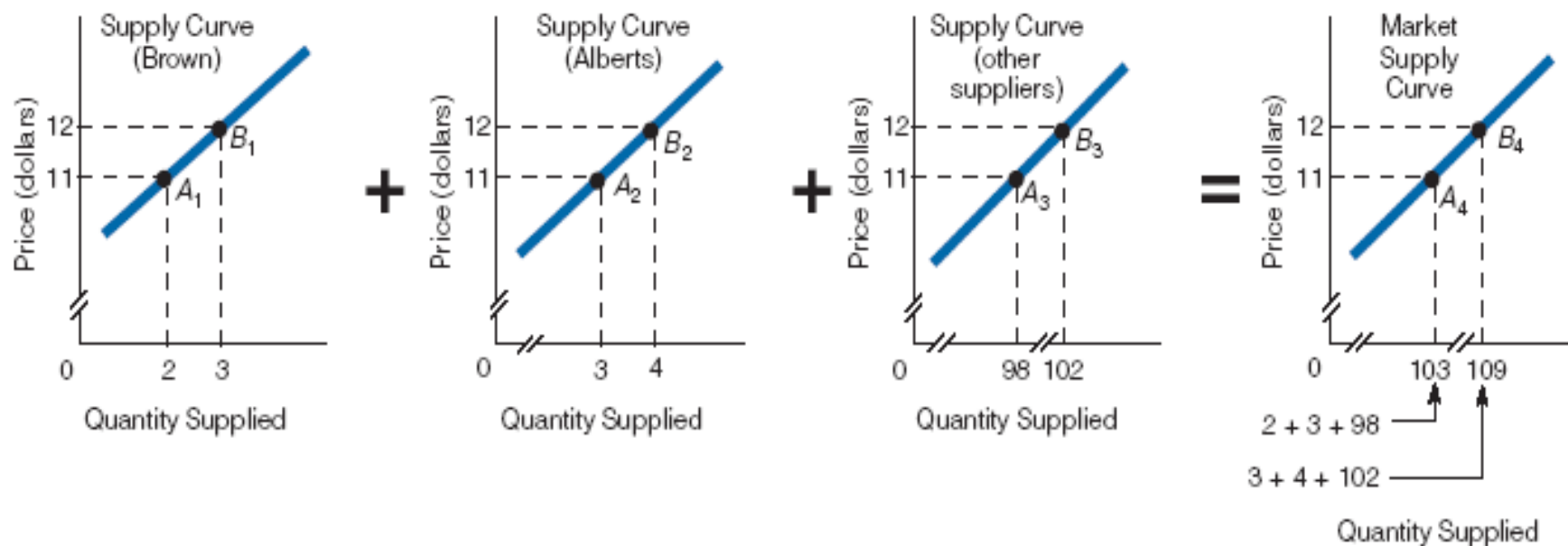
- The numerical tabulation of the quantity supplied of a good at different prices.
- A supply schedule is the numerical representation of the law of supply.

Deriving a Market Supply Schedule

Price	Quantity Supplied			
	Brown	Alberts	Other Suppliers	All Suppliers
\$10	1	2	96	99
11	2	3	98	103
12	3	4	102	109
13	4	5	106	115
14	5	6	108	119
15	6	7	110	123

(a)

Deriving a Market Supply Curve



Changes in Supply

Shift in Supply

- The supply of a good can rise or fall. An increase in the supply of a good means that suppliers are willing and able to produce and offer to sell more of the good at all prices.
- The supply of a good decreases if sellers are willing and able to produce and offer to sell less of the good at all prices.

Factors that Cause the Supply Curve to Shift

- 1) ➤ Prices of relevant resources
- 2) ➤ Technology
- 3) ➤ Prices of Other Goods
- 4) ➤ Number of sellers
- 5) ➤ Expectation of future prices
- 6) ➤ Taxes and subsidies
- 7) ➤ Government restrictions

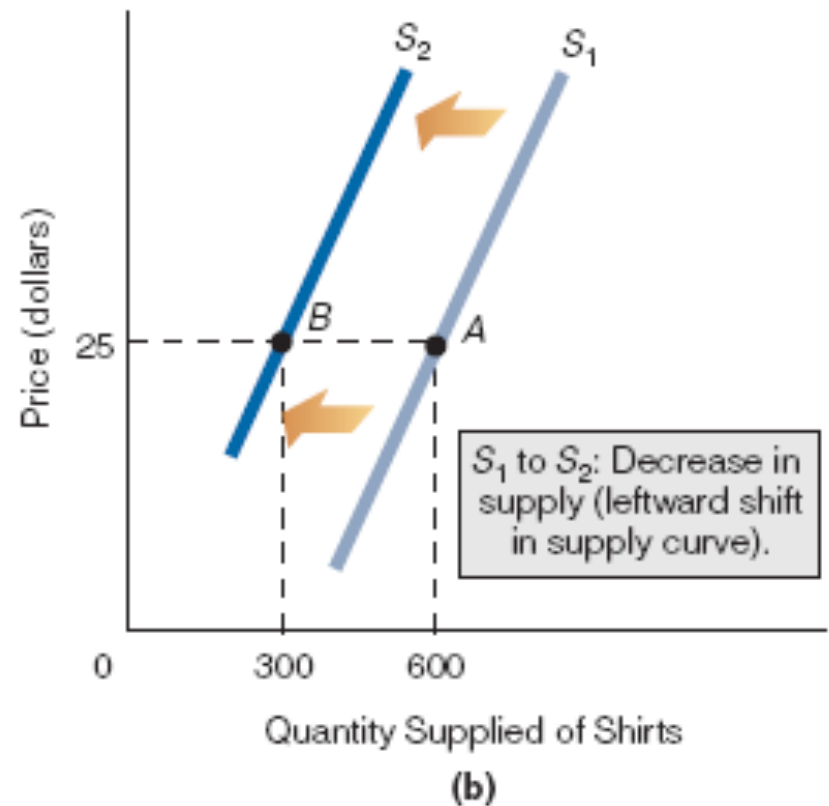
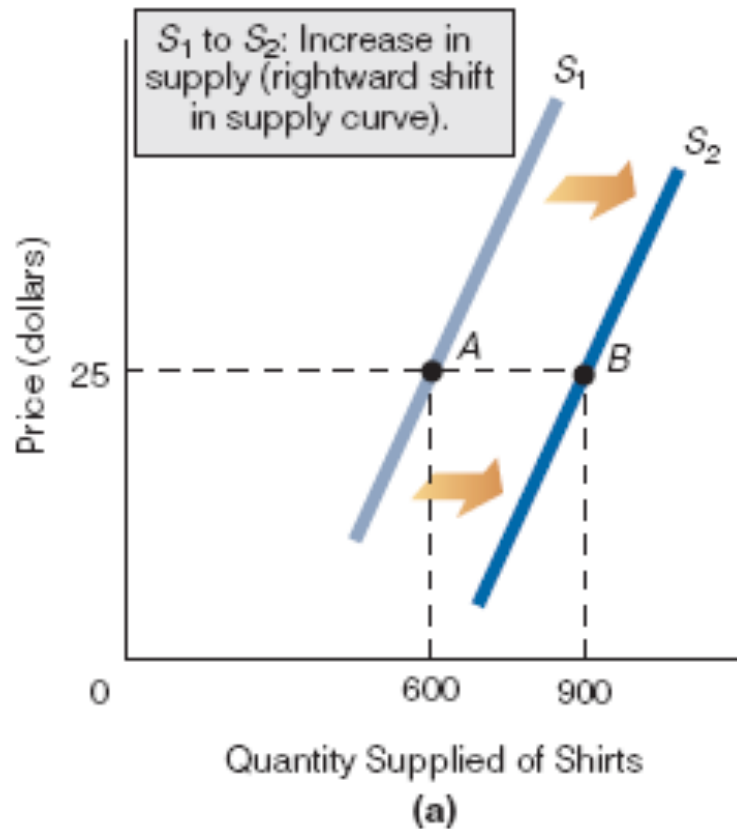
Subsidy – A Definition

- Subsidies are a monetary payment by government to a producer of a good or service
- For example, suppose the government subsidizes the production of corn by paying corn farmers \$2 for every bushel they produce. Because of the subsidy, the quantity supplied of the corn is greater at each point, and the supply curve of the corn shifts rightward. The removal of the subsidy shifts the supply curve of corn leftward.

- Bisa juga subsidi
ini utk kurangkan
pengeluaran...



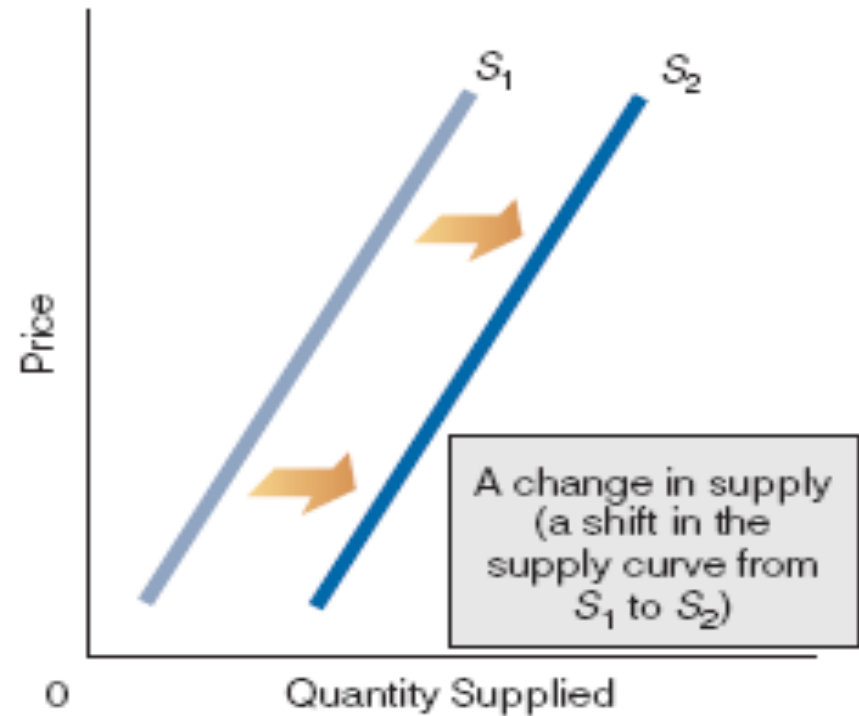
Change in Supply



Change in Supply

A Change in any of these (shift) factors can cause a change in supply

1. **Prices of Relevant Resources**
2. **Technology**
3. **Prices of other goods**
4. **Number of sellers**
5. **Expectations of Future Price**
6. **Taxes and Subsidies**
7. **Government Restrictions**

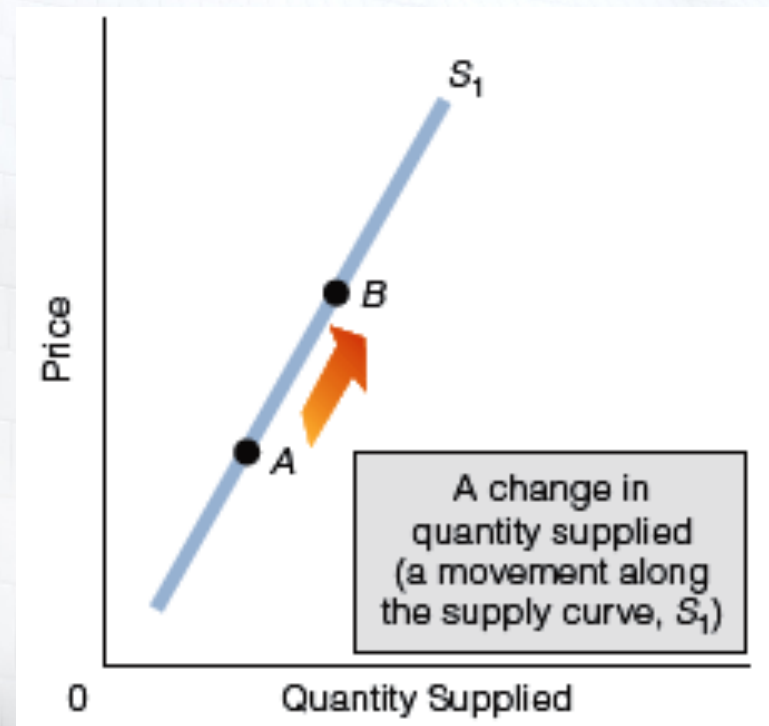


Change in Quantity Supplied

- A change in quantity supplied refers to a movement along a supply curve.
- The only factor that can directly cause a change in the quantity supplied of a good is a change in the price of the good, or own price.

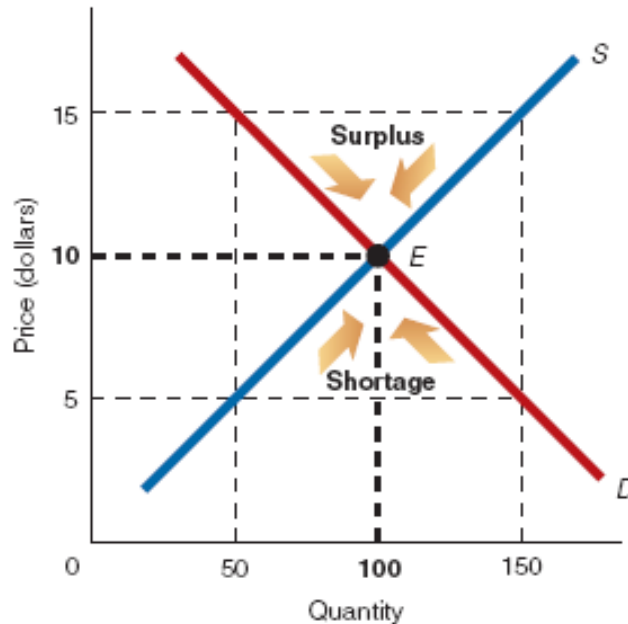
Change in Quantity Supplied

A change in this (movement) factor will cause a change in quantity supplied:
1. (A good's) own price

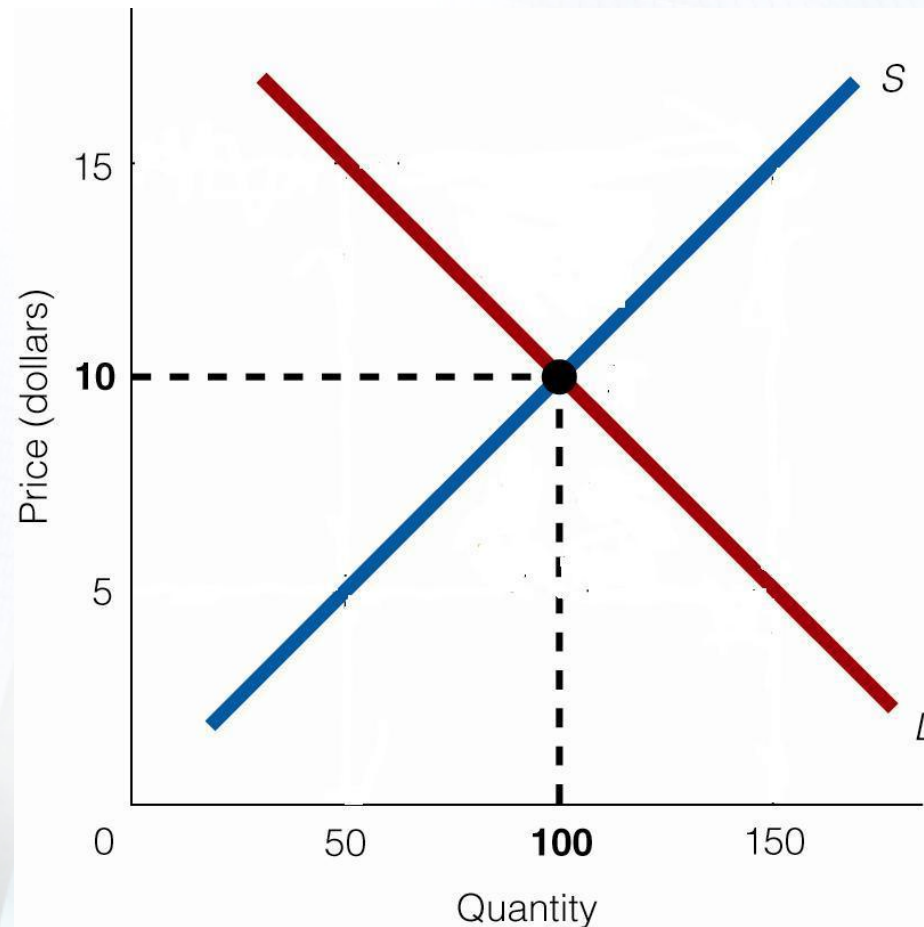


Market Equilibrium

Click graph to watch tutorial



Putting Supply and Demand Together



Market Equilibrium

→ *Stabilized*
→ *Balanced*

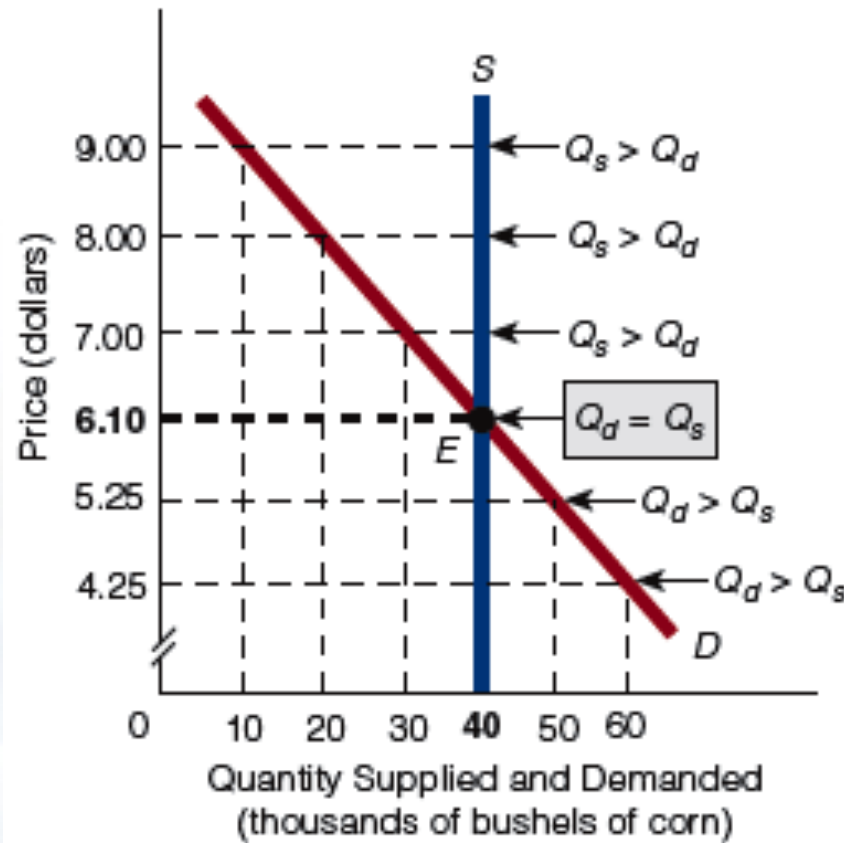
- Equilibrium in a market is the price quantity combination from which there is no tendency for buyers or sellers to move away.
- Graphically, equilibrium is the intersection point of the supply and demand curves.

Auction at Work in a Market

Supply and Demand at Work

- The auctioneer calls out different prices, and buyers record how much they are willing and able to buy.
- At prices of \$9.00, \$8.00, and \$7.00, quantity supplied is greater than quantity demanded.
- At prices of \$4.25 and \$5.25, quantity demanded is greater than quantity supplied.

Auction at Work in a Market



At a price of \$6.10, quantity demanded equals quantity supplied.

Surplus and Shortage

$$Q_S > Q_D$$

Surplus (Excess Supply) - A condition in which quantity supplied is greater than quantity demanded. Surpluses occur only at prices above equilibrium price.

Shortage (Excess Demand) - A condition in which quantity demanded is greater than quantity supplied. Shortages occur only at prices below equilibrium price.

$$Q_D > Q_S$$

Equilibrium

Equilibrium Price (Market- Clearing Price)

The price at which quantity demanded of the good equals quantity supplied

Equilibrium Quantity

The quantity that corresponds to equilibrium price. The quantity at which the amount of the good that buyers are willing and able to buy equals the amount that sellers are willing and able to sell, and both equal the amount actually bought and sold.

Disequilibrium

Disequilibrium

A state of either surplus or shortage in the market

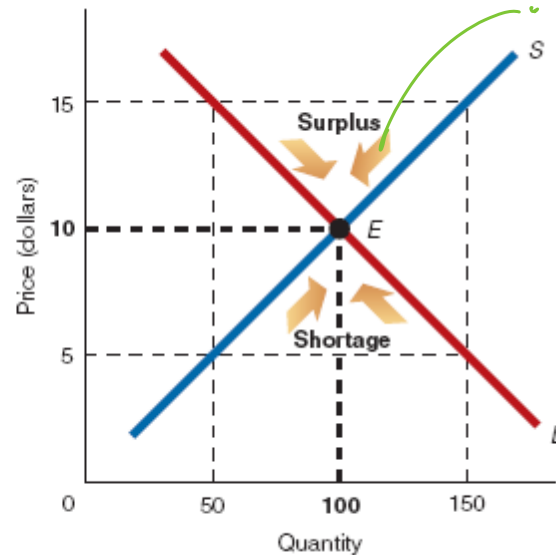
Disequilibrium Price

A price other than equilibrium price. A price at which the quantity demanded does not equal the quantity supplied.

surplus shortage

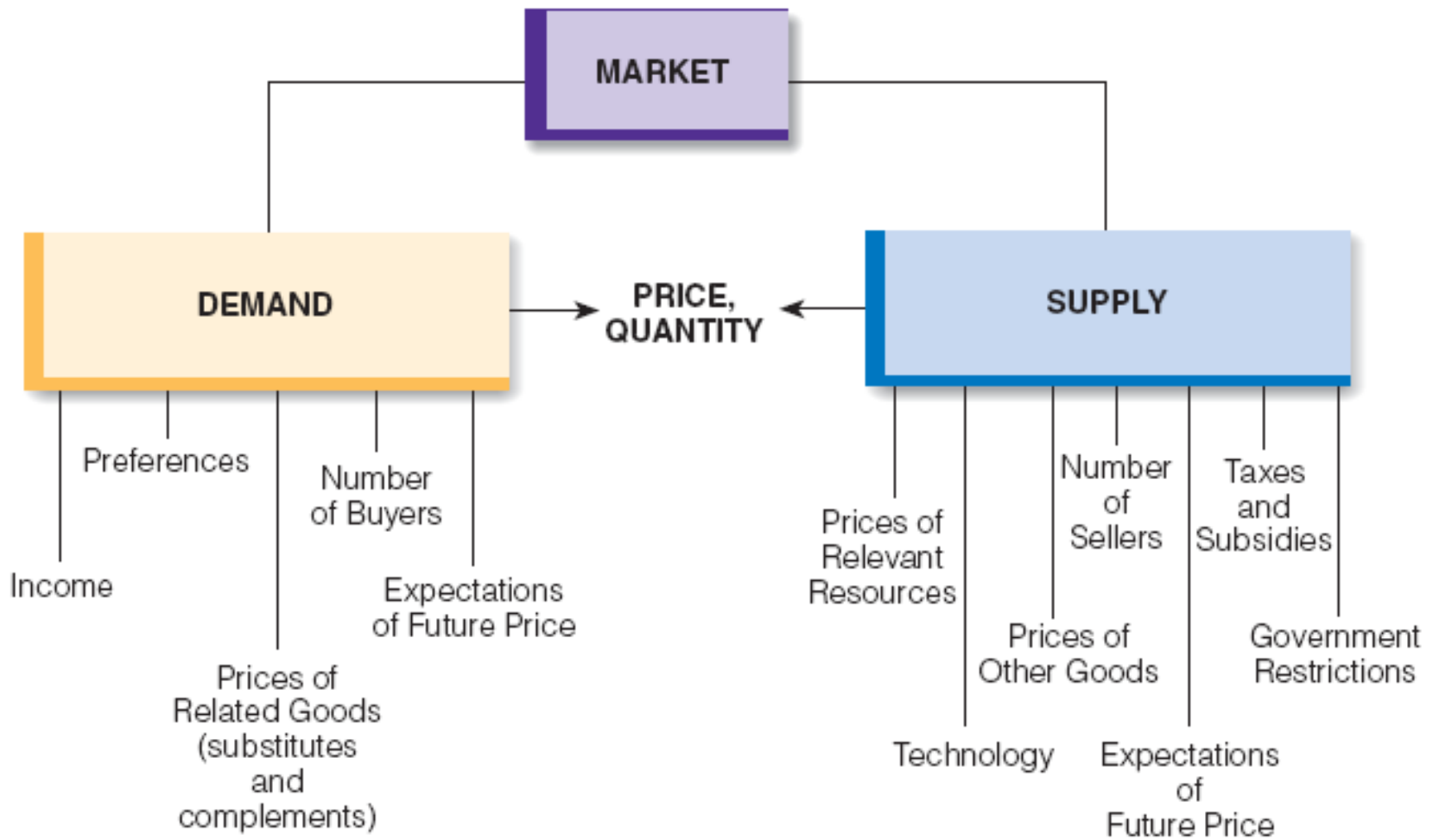
Move to Market Equilibrium

Price	Q_s	Q_d	Condition
\$15	150	50	Surplus
10	100	100	Equilibrium
5	50	150	Shortage



Force
= because people
are not willing pay
= so need to reduce
the price

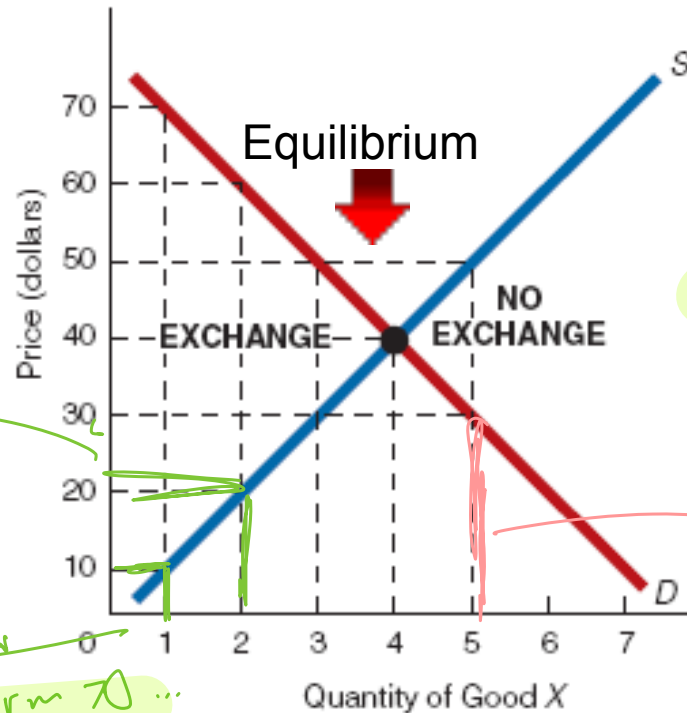
Market Demand and Supply



Moving to Equilibrium

Selaku Masih Exam !!

Units of Good X	Maximum Buying Price	Minimum Selling Price	Result
1st	\$70	\$10	Exchange
2d	60	20	Exchange
3d	50	30	Exchange
4th	40	40	Exchange
5th	30	50	No Exchange



Demand side
- Consumer willing to pay
at 1st = RM 70
then trade
xleh berlabar

dorg nak beli
pada harga 30...
tapi Firm ash
RM 50, so they
are not willing

consumer nh
beli satu...
dorg bole bayar rm 70...
Firm untung klt
(RM 10)

Consumer and Producer Surplus

area dibawah kelah permintaan
dan diatas kelah penawaran

- area dibawah
marginal
adalah total



market
equilibrium
price

consumer surplus
itu melebihi \$5

Consumer Surplus

- seronoh punya value

CS = Maximum buying price - Price paid

The difference between the maximum price a buyer is willing and able to pay for a good or service and the price actually paid.

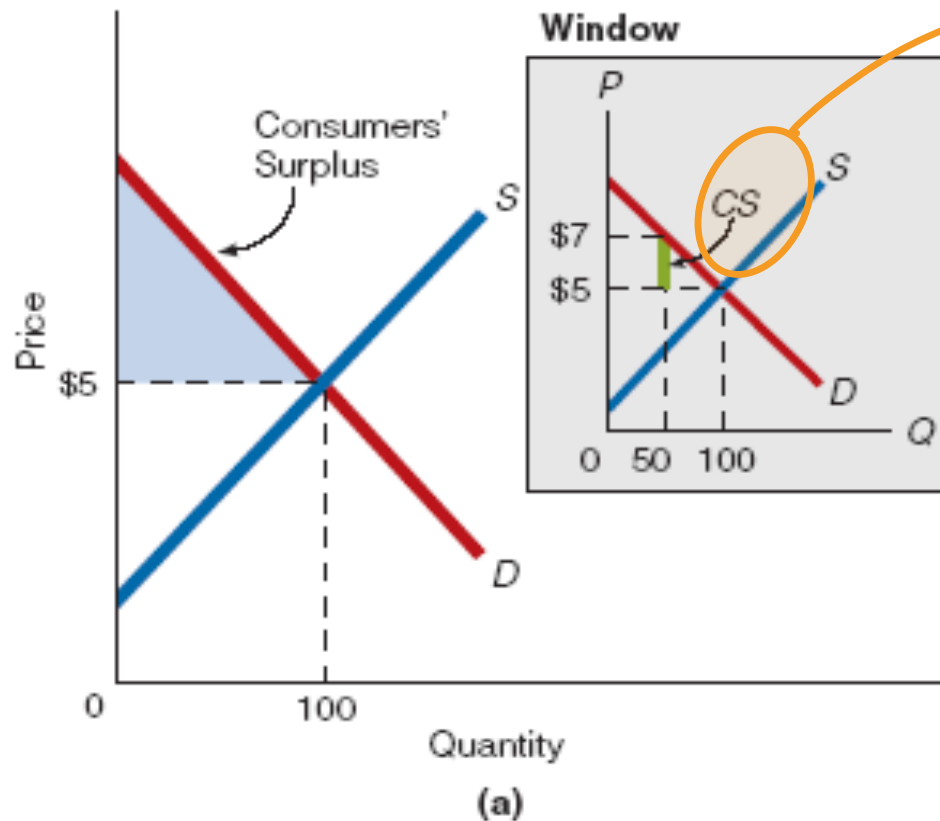
- Perbezaan yg consumer sanggup bayar dengan harga yang dibayar

☞ satisfaction

- kepuasan,
man, its
not real
money



Consumer Surplus



Consumers' Surplus (CS)

consumer surplus di
rum 2
pada
kuantitas
50

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Producer Surplus

PS = Price received - Minimum Selling Price

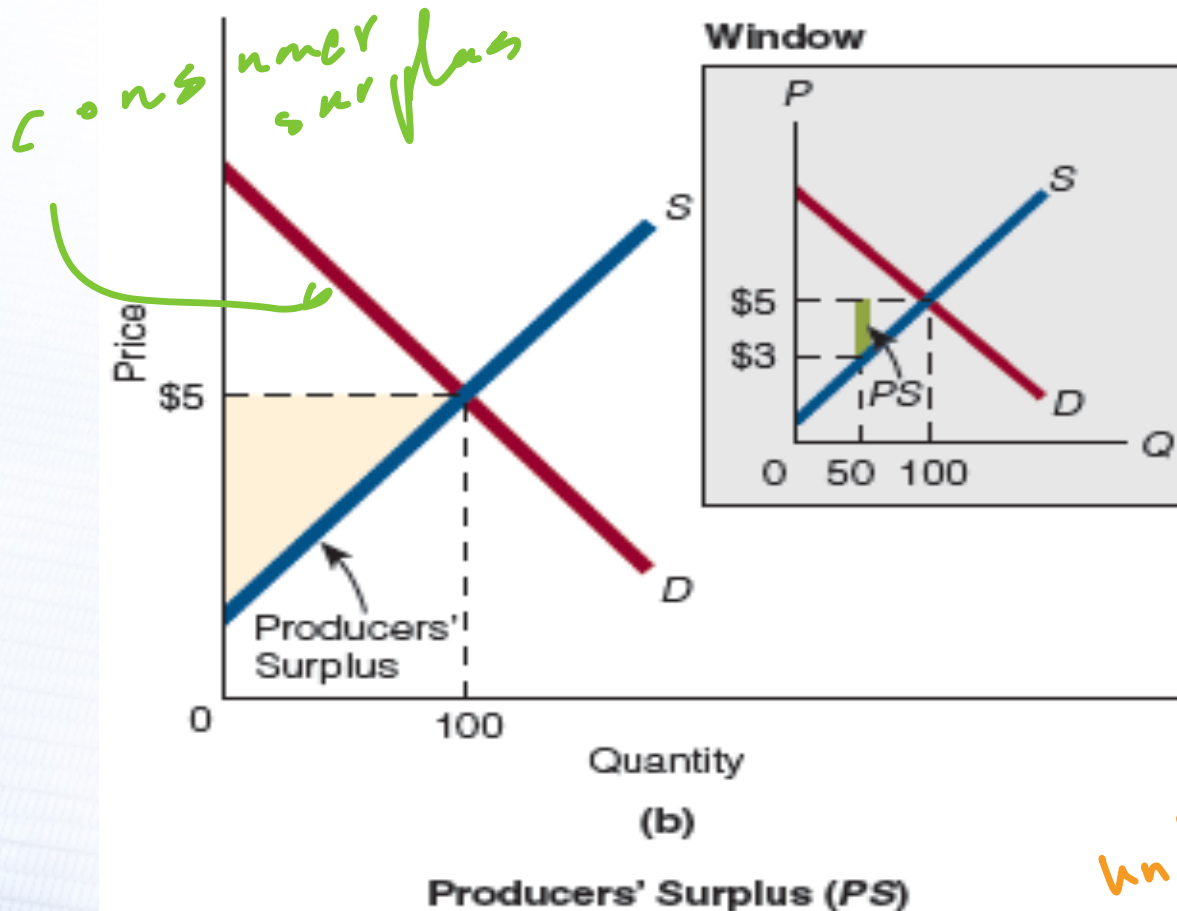
The difference between the price sellers receive for a good and the minimum or lowest price for which they would have sold the good.



Perbedaan antara
harga minimum
ataupun harga yang
diterima

Producer Surplus

- dapat dit
lebih



- Revenue
- Hasil
- Nol
- unfung
- Fokushan

hasil
untuk jawab
leat Exam

Total Surplus

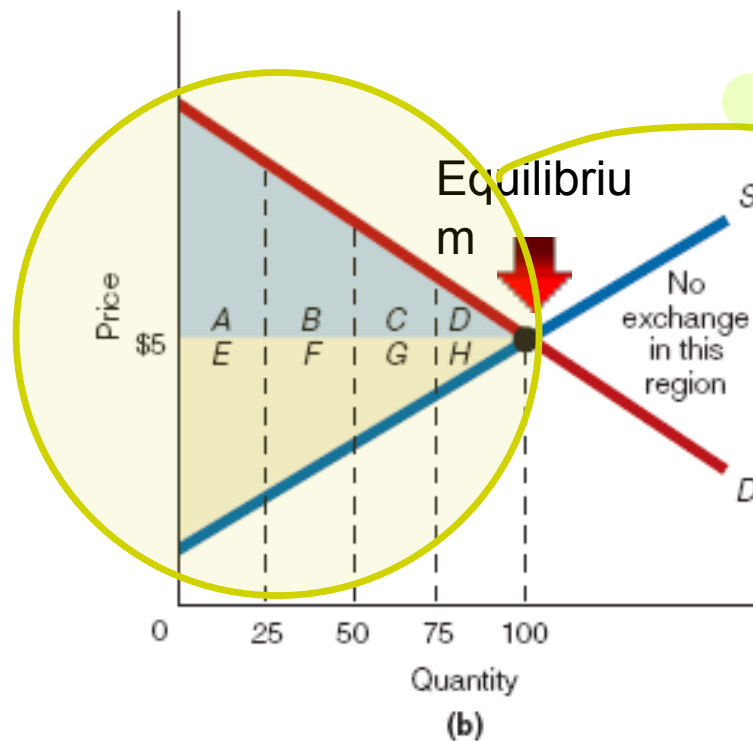
Total Surplus (TS) $TS = CS + PS$

The sum of consumers' surplus and producers' surplus.

Total Surplus

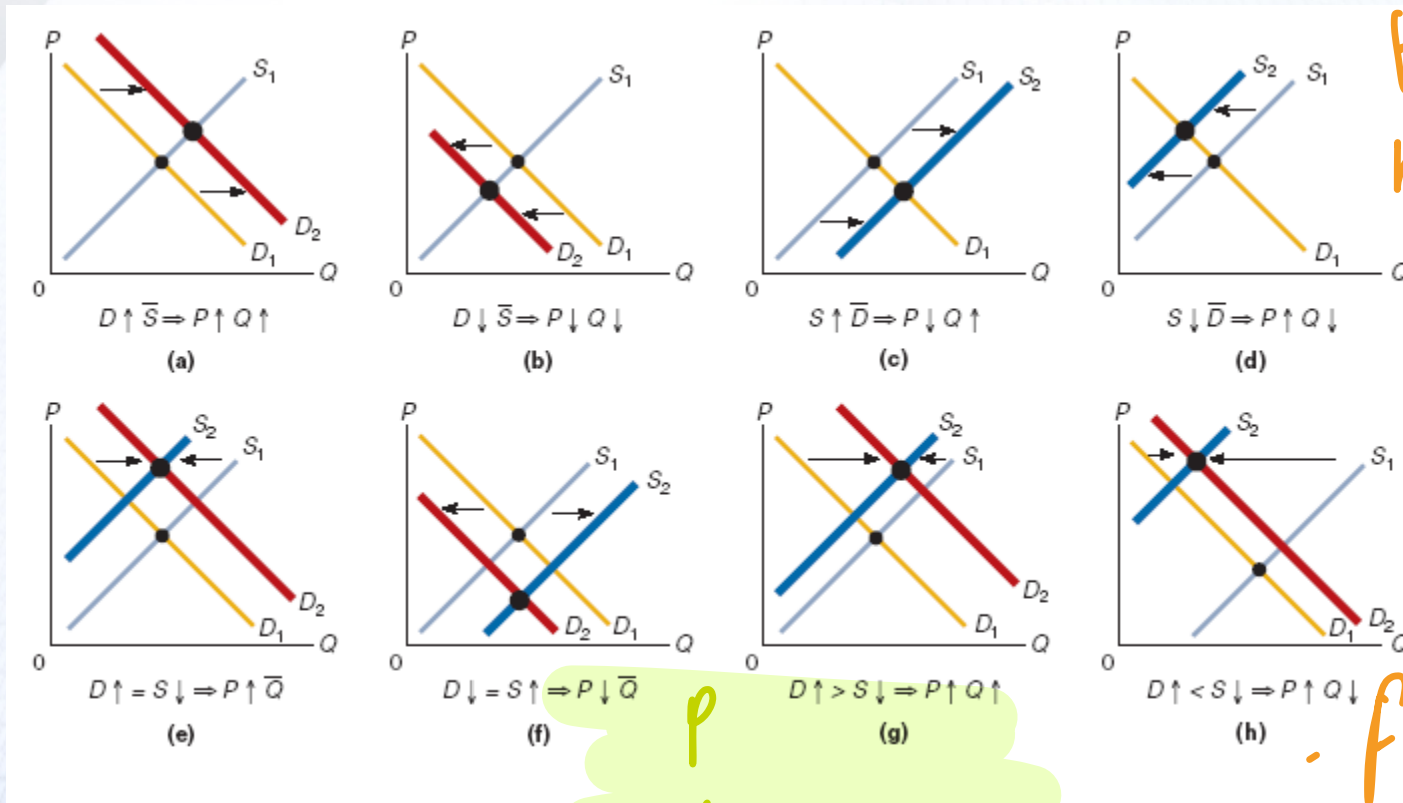
Quantity (units)	Consumers' Surplus	Producers' Surplus
25	A	E
50	A + B	E + F
75	A + B + C	E + F + G
100 (Equilibrium)	A + B + C + D	E + F + G + H

(a)



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Equilibrium Price and Quantity Effects of Supply and Demand Curve Shifts



- Dalam
Ekonomi
nyata: dia
bagi harga
and kita
kenapa ini
sendiri

. faktor ?

menyebabkan
Demand & Supply

DEMAND AND SUPPLY AS EQUATIONS

Let's now look at demand and supply as equations. Here is a demand equation: $Q_d = 1,500 - 32P$

To see what this equation says, we let price (P) *in the equation equal \$10 and then solve for quantity demanded Q_d . We get 1,180.*

$$Q_d = 1,500 - 32(10) = 1,180$$

So this equation says that if price is \$10, it follows that quantity demanded is 1,180 units. We could find other quantities demanded by plugging in different dollar amounts for price (P).

DEMAND AND SUPPLY AS EQUATIONS

Now here is a supply equation: $Q_s = 1,200 + 43P$

To find what quantity supplied (Q_s) *equals at a particular price*, we let \$5 equal price (P) and solve for quantity supplied. We get 1,415.

$$Q_s = 1,200 + 43(5) = 1,415$$

Now suppose we want to find equilibrium price and quantity given our demand and supply equations. How would we do it?

DEMAND AND SUPPLY AS EQUATIONS

First, we know that in equilibrium the quantity demanded (Q_d) of a good is equal to the quantity supplied (Q_s), so let's set the two equations equal to each other this way:

$$1,500 - 32P = 1,200 + 43P$$

Now we can solve for P . We add $32P$ to both sides of the equal sign and subtract $1,200$ from both sides. We are left with: $75P = 300$; It follows then that $P = 300/75$ or \$4.00.

DEMAND AND SUPPLY AS EQUATIONS

Once we know equilibrium price is \$4.00, we can place this value in either the demand or supply equation to find the equilibrium quantity. Let's place it in the demand equation:

$$Q_d = 1,500 - 32(4.00) = 1,372$$

Just to make sure that 1,372 is also the quantity supplied, we put the equilibrium price of \$4.00 into the supply equation:

$$Q_s = 1,200 + 43(4.00) = 1,372$$

In summary, given our demand and supply equations, equilibrium price is \$4.00 and equilibrium quantity is 1,372.